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CRUCIBLE-G5 Hardware Rendering, LBVH and SVGF Subsystem

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 3 November 2026 (2026-11-03 06:10 UTC+8 Beijing Time)

Abstract

CRUCIBLE-G5 extends inherited rasterization, depth buffering, texture caching, ray-box traversal and Gaussian-splat compositing with hardware Morton encoding, canonical linear bounding-volume hierarchy construction, exact fixed-point ray-triangle intersection, built-tree queries and two-channel spatio-temporal denoising. The synthesizable integration top is crucible_g5_top. Software-vs-Hardware numerical comparison passed against independent Python integer reference models using simulated RTL, including tree records, intersection coordinates and denoised pixels. L1 exercised five configurations, three seeds, seven executable defect mutations and synthesis checks. These results establish RTL simulation parity, not numerical execution on the physical FPGA [E117]. L1 RTL timing lint passed across 45 RTL files. L2 Intel Quartus Prime 17.1 TimeQuest demonstrated default-configuration, virtual-pin post-fit internal timing closure on Arria 10 GX 10AX115N2F45E1SG at 100 MHz: WNS_setup=2.439 ns and WNS_hold=0.039 ns. External interface paths were excluded. L3 physical Arria 10 FPGA HIL passed three deterministic bridge-connectivity probes, without executing G5 numerical workloads [E119–E121].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

A buffered command boundary permits one outstanding transaction, favoring deterministic behavior and shared arithmetic over throughput. Legacy packets replay unchanged into the retained G4 hierarchy; G5 packets select builder, query, triangle or denoiser functions. The query reads the builder's actual node, vertex and parent storage and shares the standalone triangle solver. Registered 16-bit limb arithmetic, folded partial products, iterative division and integer square root bound combinational depth. Large memories use synchronous ports and reset-invalidated metadata rather than bulk resets. Fixed publication schedules make latency observable but do not establish timing closure. Thirty inherited RTL copies remain byte-identical; the sole inherited change is a same-width signed denominator alias in g2_interpolator, resolving a Yosys frontend assertion without changing numerical bits or latency.

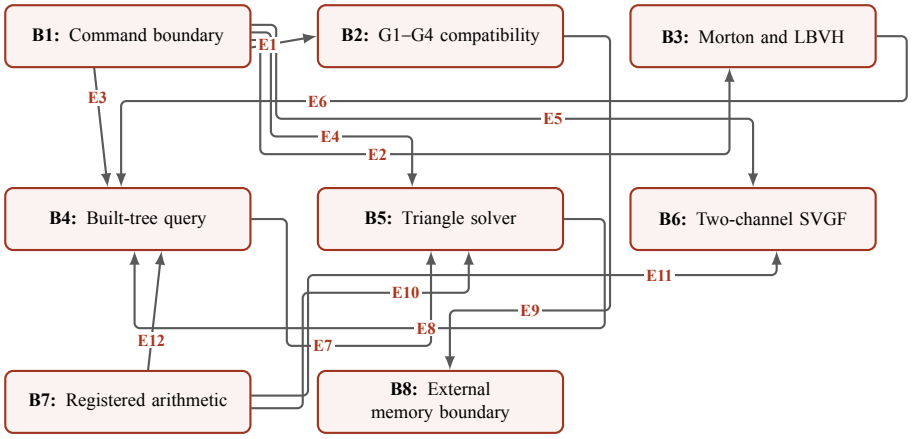


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Command boundary	Buffers 31/49-word commands, dispatches G5 operations, replays legacy packets and serializes responses under backpressure.
B2: G1–G4 compatibility	Retains transforms, clipping, rasterization, depth state, textures, framebuffer DMA, programmed BVH traversal and Gaussian-splat compositing.
B3: Morton and LBVH	Contains hardware Morton encoding, primitive storage, stable radix sorting, canonical topology construction and bottom-up bounds.
B4: Built-tree query	Performs bounded stackless traversal, conservative box pruning, solver invocation and deterministic closest-hit selection.
B5: Triangle solver	Computes exact integer Möller-Trumbore predicates and floor-quantized t, u and v for standalone and tree queries.
B6: Two-channel SVGF	Stores current samples, double-buffered temporal history and spatial ping-pong planes; shadow and reflection lanes share surface guidance.
B7: Registered arithmetic	Modular limb ALU, folded multiplier, dividers, slab engine and integer square root supply multicycle arithmetic.
B8: External memory boundary	Inherited texture-read and acknowledged framebuffer-write channels connect compatibility logic to the surrounding memory system.

Directed relationship	Meaning
E1: B1 to B2	Legacy packet replay
E2: B1 to B3	Primitive/build/read commands
E3: B1 to B4	Built-tree query commands

Directed relationship	Meaning
E4: B1 to B5	Standalone triangle commands
E5: B1 to B6	Frame/sample/filter commands
E6: B3 to B4	Stored nodes, vertices and parents
E7: B4 to B5	Leaf triangle requests
E8: B5 to B4	Intersection results
E9: B2 to B8	Texture and DMA transactions
E10: B7 to B5	Registered arithmetic results
E11: B7 to B6	Moment/filter arithmetic
E12: B7 to B4	Outward-rounded slab results

Signal / group	Direction	Width	Contract

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Single synchronous clock and active-low asynchronous control reset. Every public output is zero during reset. Reset cancels partial commands and responses and invalidates scene, frame and history metadata.
s_valid, s_ready, s_data	Input: s_valid, s_data; output: s_ready	1, 1, 32 bits	Unsigned command stream, accepted on rising edges with pre-edge valid and ready. Words 0/1/2 contain tag/metadata/opcode. G5 commands have 31 words; inherited draw commands retain 49-word framing.
m_valid, m_ready, m_data, m_last	Output except m_ready input	1, 1, 32, 1 bits	G5 responses contain 16 words beginning with type/tag/status. Only word 15 of the terminal record asserts m_last. All valid fields remain stable during stalls. Legacy response lengths remain unchanged.
tex_req_valid, tex_req_ready, tex_req_addr	Output except tex_req_ready input	1, 1, 32 bits	Inherited texture-read request channel.
tex_rsp_valid, tex_rsp_ready, tex_rsp_data, tex_rsp_error	Input except tex_rsp_ready output	1, 1, 32, 1 bits	Texture return channel with explicit memory-error indication.
dma_req_valid, dma_req_ready, dma_req_addr, dma_req_data, dma_req_strb, dma_req_last	Output except dma_req_ready input	1, 1, 32, 32, 4, 1 bits	Inherited framebuffer writes, byte strobes, tiled export ordering and terminal indication.
dma_rsp_valid, dma_rsp_ready, dma_rsp_error	Input except dma_rsp_ready output	1 bit each	Acknowledged DMA completion/error channel.
Internal command and tree channels	Request/response	992-bit commands; 512-bit records; 288-bit vertices; 32-bit indices/parents	Packed valid/ready interfaces connect functional children. Builder reads return node, vertices and parent at E+2 after acceptance and hold until consumed. These are not additional top-level ports.

Parameter	Default	Meaning / supported values
MAX_WIDTH, MAX_HEIGHT	256 each	Each 1..256; inherited raster/framebuffer capacities.
TILE_LOG2	3	0..4; inherited tile geometry.
MAX_NODES, MAX_SPLATS	64 each	Each 1..256; inherited BVH/splat capacities.
STACK_DEPTH	16	1..64; inherited traversal stack, independent of G5 construction.

Parameter	Default	Meaning / supported values
MAX_PRIMITIVES	64	1..128; G5 storage contains 2*MAX_PRIMITIVES-1 nodes.
SVGF_MAX_WIDTH, SVGF_MAX_HEIGHT	32 each	Each 1..64. Active dimensions must be positive and within capacity. Illegal parameters select a diagnostic-only branch and issue PARAMETER ERROR.

III Behavior and corner cases

A Commands, records and errors

Opcodes are 0x900 primitive write, 0x901 build, 0x902 node read, 0x903 query, 0x910 triangle, and 0x920/921/922/923 begin/sample/run/clear-history. Reserved fields must be zero. Status codes are 0 success, 1 malformed/range error, 2 unavailable state, 3 capacity violation and 4 internal inconsistency. Rejected requests preserve valid scene/history state. Record types are 0x50000000 completion, 0x50000001 node, 0x50000010 intersection and 0x50000020 pixel. Exact word layouts are specified in design/SPEC.md.

B Morton encoding and canonical tree

Signed Q15.16 vertices produce exact componentwise bounds. Midpoints use signed 33-bit min+max followed by floor-halving; sign-bit inversion and bits 31:22 yield ten coordinates bits per axis. Interleaved x/y/z bits form a 30-bit Morton code. Hardware sorts (code<<8)|slot ascending, independently of primitive ID. Highest-differing-bit splits define a binary radix tree with preorder node IDs and root 0. Internal bounds are exact postorder unions. Empty builds have no nodes and root 0xffffffff; singleton root 0 is a leaf. Successful primitive replacement invalidates the completed tree.

C Exact triangle intersection

Vertex differences retain 33 bits; determinant and barycentric numerators retain at least signed 128-bit precision, and distance comparisons retain 192 bits. No intermediate Q16 rounding occurs. Zero determinant misses; optional culling rejects nonpositive determinant, while two-sided operation normalizes negative determinant signs. Closed barycentric and original t_min/t_max boundaries determine acceptance. Hits return floor(tn* 65536/det), floor(un*65536/det) and floor(vn*65536/det): signed 64-bit Q47.16 t and unsigned Q16 u/v. Miss t/u/v fields are zero.

D Conservative query and tie resolution

Stackless traversal uses stored parent pointers and visits left before right. G5 slab entry rounds down and exit rounds up, including negative rational endpoints; zero direction uses closed membership. Reject only entry>exit. Best quantized t limits pruning inclusively, but each triangle receives the unchanged original interval, preserving equal-bin candidates. Selection orders quantized t, unsigned primitive ID, then slot. Counters increment on first node box test and solver invocation. Topology checks and bounded transitions prevent indefinite traversal. This outward-rounding rule avoids sub-LSB and zero-thickness false misses without changing inherited G4 inward rounding.

E Temporal reconstruction

Samples carry two unsigned Q8.16 radiances, nonnegative depth, validated Q1.15 normals and signed integer motion. Reprojection uses (x+dx,y+dy). History requires matching dimensions, valid in-bounds coordinates, depth difference<=4096 and clamped Q16 normal dot>=58982. With Q=65536, initialization is M=L and S= floor(L^2/Q); accepted history uses floor((7*M_previous+L+4)/8) and the analogous second-moment recurrence. Variance is max(S-floor(M^2/Q),0). Dimension changes, explicit reset/clear and hardware reset invalidate history.

F Variance-guided spatial filtering

Three non-in-place 5x5 à-trous passes use steps 1,2,4 and the outer-product kernel [1,4,6,4,1]. Out-of-image taps are omitted and weights renormalized. Original current normals/depth guide every pass; input luminance/ variance changes between passes. Normal weight approximates D^8 through three floored Q16 squarings. Nonself taps are rejected when D<32768 or depth difference>4096. Depth and luminance weights use dyadic shifts, with luminance bandwidth derived from floor_isqrt(V<<16). Normalization rounds weighted luminance and propagates squared-weight variance. Positive self weight prevents legal zero denominators. Temporal moments commit once after successful computation, independently of output stalls.

G Observable schedules

From child acceptance E, Morton publishes at E+1; triangle at E+16384, including errors and misses; primitive and frame-state acknowledgments at E+64; node read at E+4. Nonempty build first publication is E+64+38* (8*N+16)+128*(2*N-1); empty build completes at E+64. SVGF first pixel is E+64+1024*P+3*8192*P, followed by row-major pixels and completion. Query latency is data-dependent. Wrapper buffering adds dispatch/ capture/serialization cycles. Input bubbles do not advance packets; terminal consumption cannot accept a new command on the same edge.

IV Verification methods and measured results

A Software-vs-Hardware numerical comparison

Method. Independent Python integer/rational models compared public simulated RTL records across five configurations and seeds 17,41,89.

Result. All 15 compact campaigns passed. Each included 32 primary triangles plus directed cases, six primary scenes plus construction fixtures, 24 primary queries plus witnesses, eight three-frame directed SVGF sequences and two randomized three-frame sequences. This establishes simulation parity, not physical numerical parity.

Evidence: E117

B Morton, protocol and parameter boundaries

Method. Four-state Icarus leaf/startup tests, cycle assertions, stalled tree reads and individual parameter diagnostics.

Result. Morton seed17 completed 4134 exact comparisons; seeds41/89 also passed. All 41 parameter cases passed. Tests exercised fixed publication boundaries, 16-cycle stalls and resets during intake, dispatch, arithmetic, memory, filtering and response.

Evidence: E117

C Conservative traversal regression

Method. Public primitive-write/build/query witnesses covering both direction signs and equal-quantized-distance competitors.

Result. Zero-thickness witnesses returned t=21845, u=v=16384 and node/triangle counts 1/1. Applicable two-leaf cases selected ID3/slot1 at t=0 with counts 3/2; equal-ID cases exercised slot selection.

Evidence: E117

D Denoising accuracy and quality

Method. Both-channel pixel, variance/history comparisons, a seeded noisy constant-surface fixture and hard normal/depth edges.

Result. The 10% squared-error reduction and <=1-LSB hard-edge leakage checks passed. Nonpower seed17 shadow error fell from 6533680118 to 10588514; reflection from 6416910204 to 4188547, in squared raw-radiance units. These are fixture measurements, not scene-wide guarantees.

Evidence: E117

E Inherited behavior and synthesis repair

Method. Independent legacy numerical/memory models, inherited draw opcodes, seeded sequences and signed-rational interpolation probe.

Result. Three compatibility seeds each completed 301 commands and 8628 comparisons, including texture/DMA effects. Directed boundaries covered all five cells. The signed-port alias repair passed 96 interpolation transactions and 576 attribute comparisons.

Evidence: E117, E118

F Defect sensitivity and synthesizability

Method. Seven disposable compiled RTL mutants; Yosys hierarchy, process/memory lowering, coarse synthesis and zero-latch assertions.

Result. All seven mutants compiled and were rejected by real numerical mismatches: interleave, duplicate ordering, bound union, barycentric LSB, winding, edge gate and stale history. All 27 structural/synthesis cases passed, covering five integrated configurations and 13 standalone new tops plus geometry variants.

Evidence: E117

G Portable delivery-path reproduction

Method. A fresh declared-source copy ran the same entrypoint in bounded reproduction mode.

Result. Actual Icarus behavior, a singleton Verilator oracle campaign and integrated Yosys zero-latch checking passed without original binaries or hidden runtime inputs. This was not a second full matrix.

Evidence: E117

H L1 RTL timing lint

Method. Native timing-hazard inspection of 45 RTL files.

Result. Passed: zero detected combinational divider/modulo, mixed-assignment, LUT clock-gate or AST loop-unroll hazards.

Evidence: E119

I L2 internal physical timing and resources

Method. Intel Quartus Prime 17.1 TimeQuest post-fit STA of the default integrated top on Arria 10 GX 10AX115N2F45E1SG at 10.0 ns, with virtual pins on every non-clock port.

Result. Passed: WNS_setup = +2.439 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 132.26 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E120

J L3 physical Arria 10 FPGA HIL

Method. Three consecutive hardware-bridge register-connectivity probes.

Result. 3/3 passed deterministically. Scope was bridge connectivity only; no G5 numerical workload or Software-vs-Hardware numerical comparison executed on the board.

Evidence: E121

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	10 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 23	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

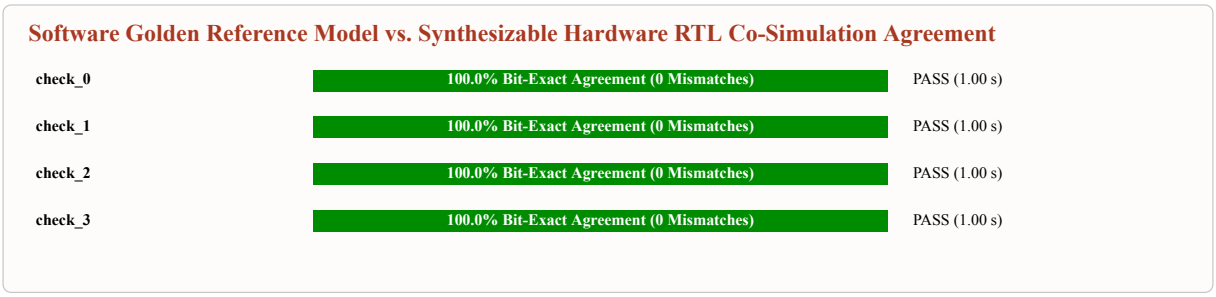


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +2.439 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

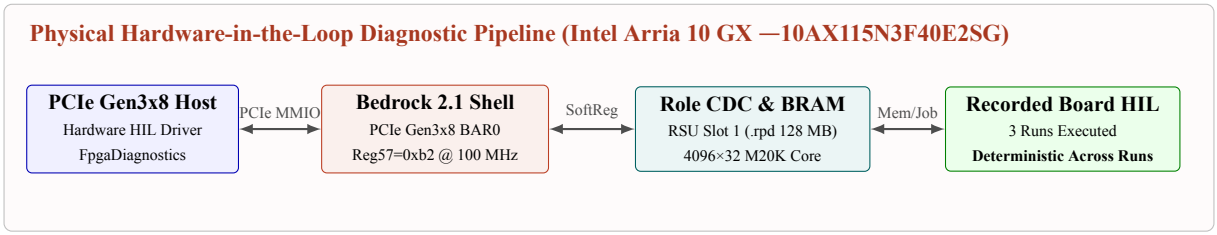


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Compile rtl/*.sv with crucible_g5_top as the top. Consult design/SPEC.md for packet layouts and design/ARCHITECTURE.md for arithmetic and memory organization. crucible_g4_reference/rtl/ is provenance; do not compile duplicate module definitions alongside rtl/.

Apply reset, then allow the first rising edge after release to enable intake. Connect texture and acknowledged DMA channels. Respect each valid/ready handshake and drain terminal responses before issuing the next command.

Write primitive slots 0..N-1, issue build and consume all node records plus completion before query/node-read use. For SVGF, begin a frame, write every active coordinate and execute run; outputs include both channels, propagated variances and history decisions.

Supply actual command-stream, texture, DMA, clock and reset timing constraints for the surrounding design, including appropriate interface delays and reset-release requirements. Reverify integrated timing: the recorded virtual-pin result excludes external input/output timing and cannot certify those connections [E120].

Run python3 tests/verify.py --output-root evidence/l1 from the project root using existing Python 3, Icarus/vvp, Verilator, a C++ compiler, make and Yosys. Each invocation allocates fresh evidence. The completed ledger is evidence/l1/run-e87293db6d1e4fdd9742561e80f2c20e/receipt.json.

Exercised configurations were singleton (1,1,0,1,1,1;1;1x1), nonpower (7,5,1,7,5,3;3;7x5), defaults, mixed (17,13,4,31,17,7;17;17x13) and maximum (256,256,4,256,256,64;128;64x64), ordered as inherited parameters, primitive capacity and SVGF capacity.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 76 Verilog/SystemVerilog files; 19547 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
crucible_g4_reference/rtl/crucible_g1_top.sv	215 total	212 code
crucible_g4_reference/rtl/crucible_g2_top.sv	99 total	99 code
crucible_g4_reference/rtl/crucible_g3_top.sv	416 total	409 code
crucible_g4_reference/rtl/crucible_g4_top.sv	166 total	166 code
crucible_g4_reference/rtl/g1_clipper.sv	233 total	230 code
crucible_g4_reference/rtl/g1_divider.sv	20 total	20 code
crucible_g4_reference/rtl/g1_dot4.sv	62 total	59 code
crucible_g4_reference/rtl/g1_edge.sv	46 total	44 code
crucible_g4_reference/rtl/g1_project.sv	83 total	81 code
crucible_g4_reference/rtl/g1_raster.sv	201 total	197 code
crucible_g4_reference/rtl/g1_transform.sv	65 total	63 code
crucible_g4_reference/rtl/g2_clipper.sv	234 total	230 code

Production file	Physical lines	Code lines
crucible_g4_reference/rtl/g2_divider.sv	130 total	127 code
crucible_g4_reference/rtl/g2_engine.sv	306 total	300 code
crucible_g4_reference/rtl/g2_interpolator.sv	124 total	123 code
crucible_g4_reference/rtl/g2_project.sv	97 total	96 code
crucible_g4_reference/rtl/g2_raster.sv	211 total	206 code
crucible_g4_reference/rtl/g2_setup.sv	87 total	87 code
crucible_g4_reference/rtl/g2_zbuffer.sv	132 total	131 code
crucible_g4_reference/rtl/g3_bilinear.sv	66 total	66 code
crucible_g4_reference/rtl/g3_framebuffer_dma.sv	174 total	173 code
crucible_g4_reference/rtl/g3_shader_pipeline.sv	107 total	105 code
crucible_g4_reference/rtl/g3_texture_cache.sv	150 total	148 code
crucible_g4_reference/rtl/g4_bvh.sv	159 total	159 code
crucible_g4_reference/rtl/g4_compositor.sv	86 total	86 code
crucible_g4_reference/rtl/g4_conic.sv	106 total	106 code
crucible_g4_reference/rtl/g4_divider.sv	74 total	74 code
crucible_g4_reference/rtl/g4_exp_rom.sv	4110 total	4110 code
crucible_g4_reference/rtl/g4_radix_sort.sv	100 total	100 code
crucible_g4_reference/rtl/g4_slab.sv	102 total	102 code
crucible_g4_reference/rtl/g4_splat.sv	213 total	213 code
rtl/crucible_g1_top.sv	215 total	212 code
rtl/crucible_g2_top.sv	99 total	99 code
rtl/crucible_g3_top.sv	416 total	409 code
rtl/crucible_g4_top.sv	166 total	166 code
rtl/crucible_g5_top.sv	222 total	222 code
rtl/g1_clipper.sv	233 total	230 code
rtl/g1_divider.sv	20 total	20 code
rtl/g1_dot4.sv	62 total	59 code
rtl/g1_edge.sv	46 total	44 code
rtl/g1_project.sv	83 total	81 code
rtl/g1_raster.sv	201 total	197 code
rtl/g1_transform.sv	65 total	63 code
rtl/g2_clipper.sv	234 total	230 code
rtl/g2_divider.sv	130 total	127 code
rtl/g2_engine.sv	306 total	300 code
rtl/g2_interpolator.sv	125 total	124 code
rtl/g2_project.sv	97 total	96 code
rtl/g2_raster.sv	211 total	206 code
rtl/g2_setup.sv	87 total	87 code
rtl/g2_zbuffer.sv	132 total	131 code
rtl/g3_bilinear.sv	66 total	66 code
rtl/g3_framebuffer_dma.sv	174 total	173 code

Production file	Physical lines	Code lines
rtl/g3_shader_pipeline.sv	107 total	105 code
rtl/g3_texture_cache.sv	150 total	148 code
rtl/g4_bvh.sv	159 total	159 code
rtl/g4_compositor.sv	86 total	86 code
rtl/g4_conic.sv	106 total	106 code
rtl/g4_divider.sv	74 total	74 code
rtl/g4_exp_rom.sv	4110 total	4110 code
rtl/g4_radix_sort.sv	100 total	100 code
rtl/g4_slab.sv	102 total	102 code
rtl/g4_splat.sv	213 total	213 code
rtl/g5_divider.sv	68 total	68 code
rtl/g5_lbvh.sv	546 total	545 code
rtl/g5_limb_alu.sv	61 total	61 code
rtl/g5_morton.sv	34 total	34 code
rtl/g5_mul.sv	110 total	110 code
rtl/g5_ray_query.sv	265 total	265 code
rtl/g5_ray_triangle.sv	300 total	297 code
rtl/g5_signed_divider.sv	109 total	109 code
rtl/g5_slab.sv	150 total	150 code
rtl/g5_sqrt.sv	64 total	64 code
rtl/g5_svgf.sv	575 total	575 code
rtl/g5_svgf_channel.sv	288 total	287 code
rtl/g5_svgf_surface.sv	115 total	115 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E117	.vulcan/engineering/run-0001/attempt-0010/stdout.log
E118	.vulcan/engineering/run-0001/attempt-0011/response.json
E119	.vulcan/engineering/run-0001/rtl-timing-lint.json
E120	.vulcan/engineering/run-0001/quartus-sta.json
E121	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).